

Blood cells classification

1. Rendering Report

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1. Introduction

Morphological analysis remains a starting cornerstone for hematological diagnostics, providing critical insights for detecting hematological malignancy, and other blood related disorders. However, this process is inherently time consuming, subject to inter observer variability, and requires specialized expertise that may be unavailable in resource constrained settings. Studies have documented significant disagreement rates even among experienced hematologists when classifying morphologically ambiguous cells, particularly in distinguishing blast populations or atypical lymphocytes [Matek et al., 2019].

The objective of this project is to design and develop an automatic blood cell type classification system.

2. Images datasets

This project integrates two complementary blood cell image datasets that together provide comprehensive coverage of both normal peripheral blood morphology and pathological cells associated with acute myeloid leukemia (AML).

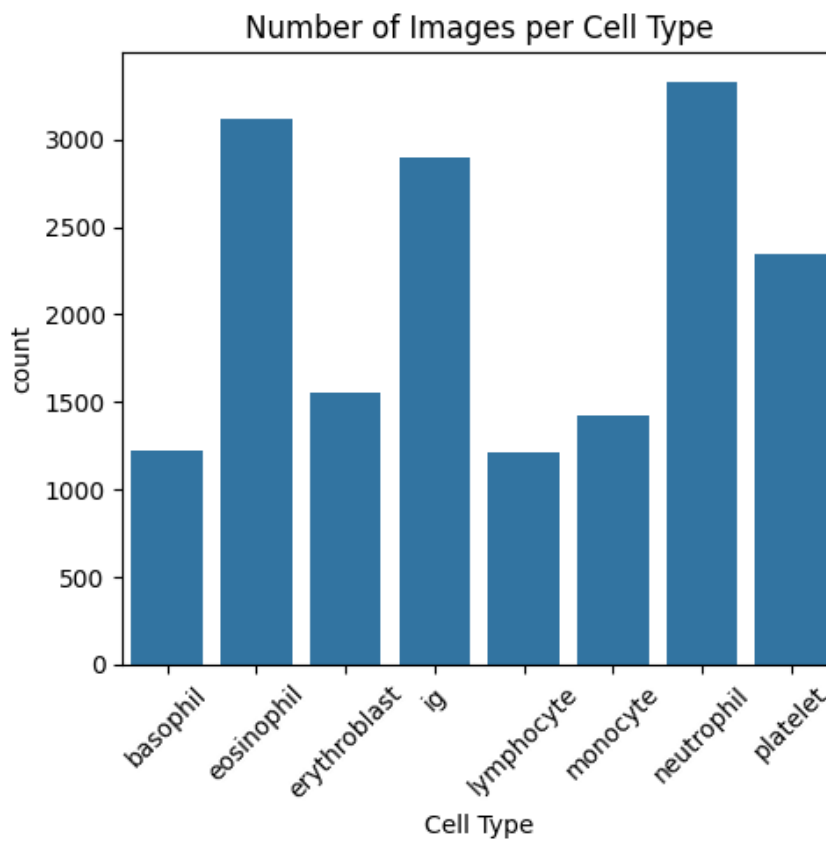
Combining these datasets provides balanced representation across the morphological spectrum required for clinical blood cell classification: The Mendeley dataset supplies robust representation of normal cell types essential for differential diagnosis. The TCIA AML dataset contributes pathological blast populations and rare immature precursors critical for leukemia detection. Together they enable training of models capable of distinguishing normal variants from malignant cells while recognizing ambiguous cases requiring expert review (e.g., atypical lymphocytes vs. blasts).

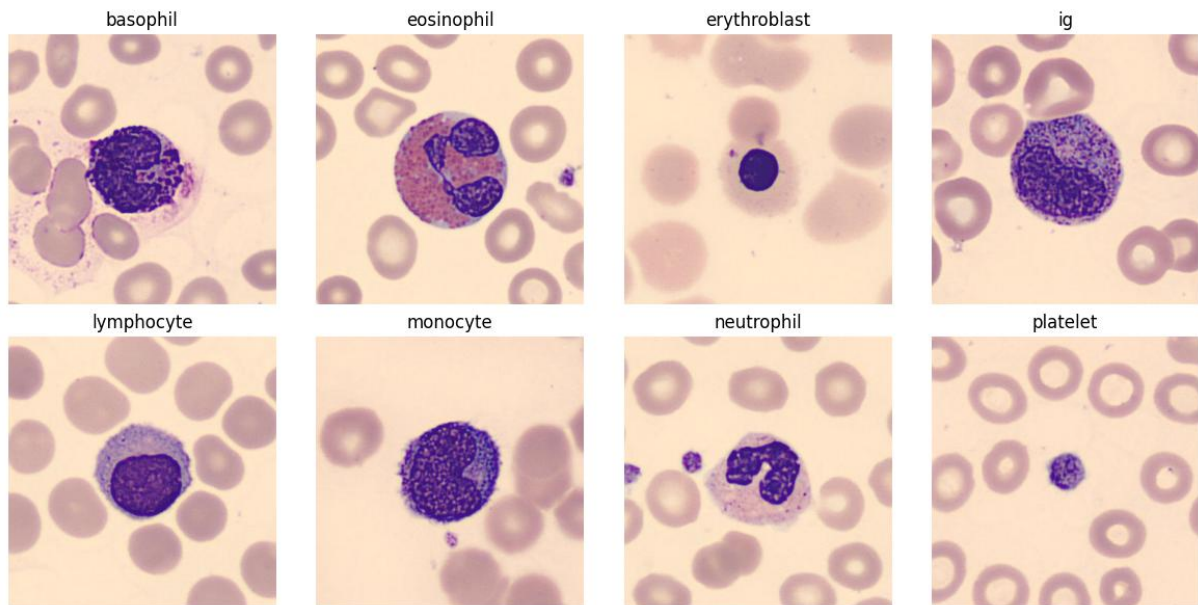
2.1 Mendeley Blood Cell Dataset

The first dataset comprises 17,092 microscopic images of normal peripheral blood cells obtained from the Hospital Clínic de Barcelona's Core Laboratory [Acevedo et al., 2019]. All images were captured using the CellaVision DM96 digital hematology analyzer at a resolution of 360×363 pixels in RGB JPG format. Blood samples were collected from donors without infections, hematologic or oncologic diseases, and pharmaceutical treatments at the time of acquisition. Smears were stained using the standardized May-Grünwald-Giemsa protocol, ensuring consistent morphological visualization across samples. The dataset encompasses eight morphological classes representing normal leukocyte populations and precursors:

1. Neutrophils (segmented and band forms)
2. Eosinophils
3. Immature granulocytes (metamyelocytes, myelocytes, and promyelocytes)
4. Platelet
5. Erythroblasts
6. Monocytes
7. Basophils
8. Lymphocytes (typical)

Index	Cell Type	count	percentage
0	Neutrophil	3329	19.48
1	Eosinophil	3117	18.24
2	ig	2896	16.94
3	Platelet	2348	13.74
4	Erythroblast	1551	9.07
5	Monocyte	1420	8.31
6	Basophil	1218	7.13
7	Lymphocyte	1214	7.10





2.1.1 Image Resolution Distribution

The analysis of image resolutions shows that the dataset contains a variety of image sizes. Where the images have many different width–height combinations and most are around 360×360, but not all.

This variety in images sizes means that some pre-processing is necessary. Therefore, it is recommended to resize all images to a standard resolution while keeping their aspect ratios, to maintain consistency and avoid any distortion. Moreover, considering the image resolution statistics:

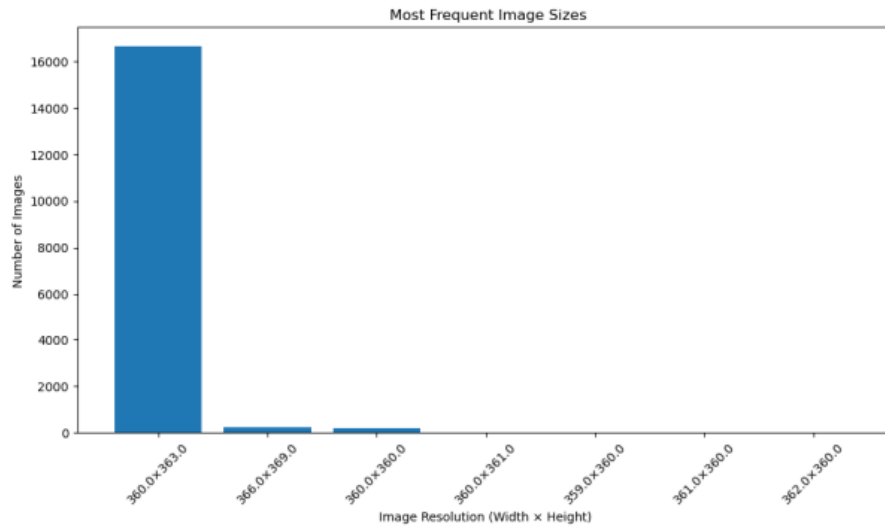
Mean width: 360.09 pixels

Mean height: 363.05 pixels

Standard deviation of width: 0.72 pixels

Standard deviation of height: 0.79 pixels

We confirm that the average image resolution was found to be 360.09 × 363.05 pixels. The standard deviations of width and height were 0.72 pixels and 0.79 pixels, respectively, indicating extremely low variability in image dimensions. This suggests that the images are nearly uniform in size. Consequently, resizing all images to a fixed resolution is statistically justified and is unlikely to introduce geometric distortion or loss of relevant cellular features.



Width	Height	Count
360.0	363.0	16639
366.0	369.0	250
360.0	360.0	198
360.0	361.0	2
359.0	360.0	1
361.0	360.0	1
362.0	360.0	1

2.1.2 Data Quality Assessment

The dataset is arranged so that each folder represents a different types of blood cells and that all images inside a folder share the same label. This structure ensures that the dataset is fully labeled and suitable for supervised learning.

A detailed data quality check was carried out, leading to the following observations:

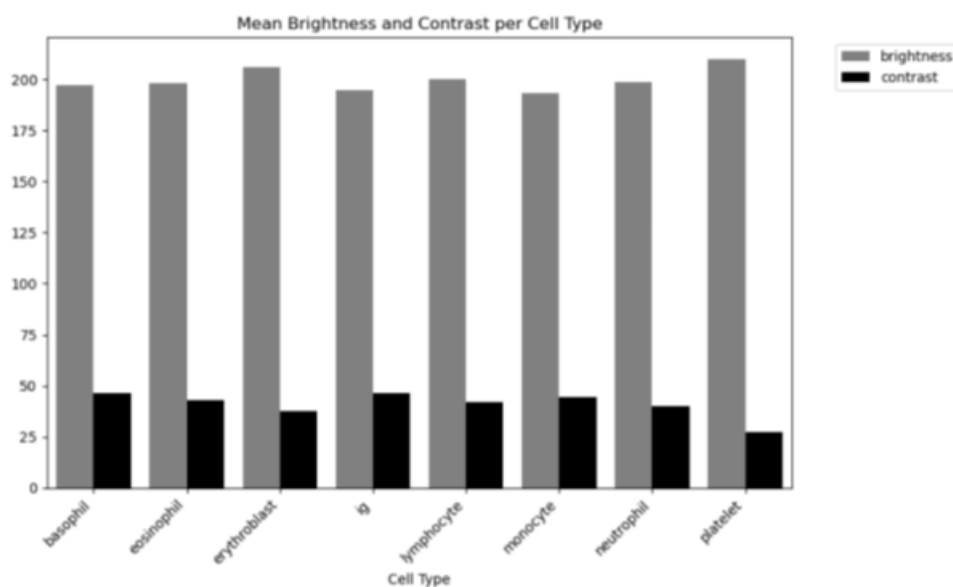
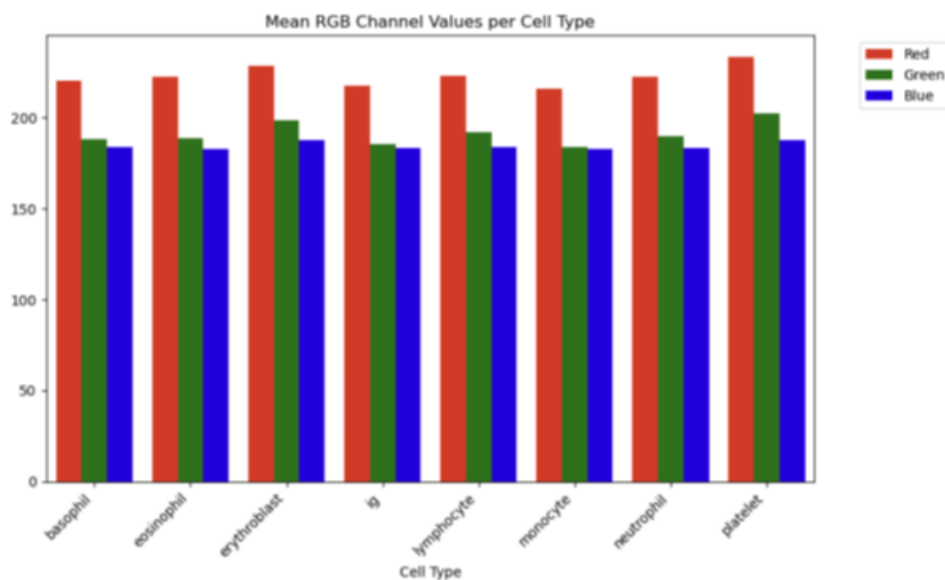
- No missing or incorrect labels were found.
- All image files paths were valid and accessible.
- No corrupted image files were detected.
- Only one image could not be read during resolution analysis and was removed from the dataset.

Overall, the dataset is clean and well-organized.

2.1.3 RGB and Grayscale Intensity Analysis

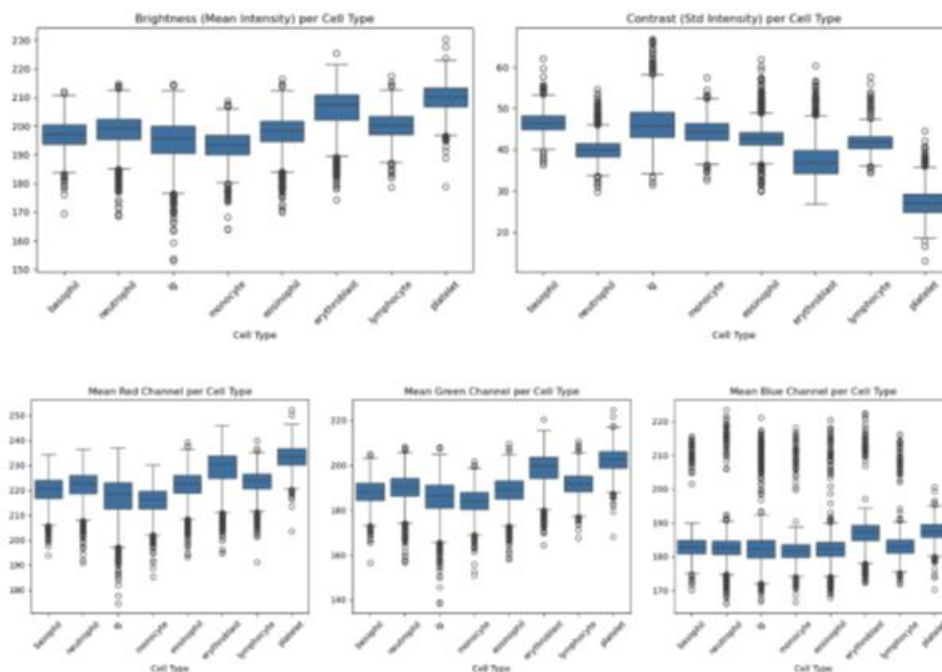
To understand the characteristics of different cell types in the datasets, we perform fundamental image analysis based on intensity and color distributions. Grayscale analysis provides insights into overall brightness and contrast patterns by converting RGB images to luminance values using the standard ITU-R BT.601 weighting ($0.299R + 0.587G + 0.114B$).

Channel-specific RGB statistics reveal color variations that may be associated with different staining properties or morphological features of cell types. These basic metrics serve as a foundation for quality control, helping to identify potential imaging inconsistencies, batch effects, or significant distributional differences between classes that could inform preprocessing decisions for the CNN classifier.



The mean RGB channel values reveal relatively consistent color profiles, with all classes showing red channel dominance (approximately 215-235) followed by green (185-202) and blue (183-190) channels. Erythroblast and platelet exhibit the highest red channel intensities (~232 and ~235 respectively), while ig displays the lowest overall intensity across all channels. The grayscale analysis demonstrates relatively uniform brightness levels (193-209) across most cell types. Platelet shows the highest mean brightness (~209) with distinctly lower contrast (~27), while ig, monocyte and basophil show slightly reduced brightness (~195-197) with higher contrast values (~47-48). Overall, the relatively narrow range of color and brightness values suggests that the CNN classifier will likely rely primarily on texture and morphological features rather than color information for differentiation.

2.1.4 Distribution Analysis with Boxplots



The boxplot visualizations show that most cell types display relatively compact interquartile ranges with medians around 195-210, though all classes exhibit outliers. Ig displays numerous low-intensity outliers extending down to ~155 and exceptionally high contrast variability with outliers extending above 60. Platelet demonstrates the narrowest contrast distribution with a median around 27, confirming its more homogeneous internal structure. For the RGB channels, ig exhibits numerous outliers spanning from ~175 to above 250 in the red channel, while platelet shows elevated green median values (~202). The prevalence of outliers across all metrics suggests some degree of imaging variability that could potentially be addressed through outlier filtering or normalization strategies during preprocessing.

The boxplot analysis indicates that the dataset is suitable for CNN training, with most cell types displaying compact interquartile ranges and consistent median values. While outliers are present across all classes, they display consistent directionality across cell types, with

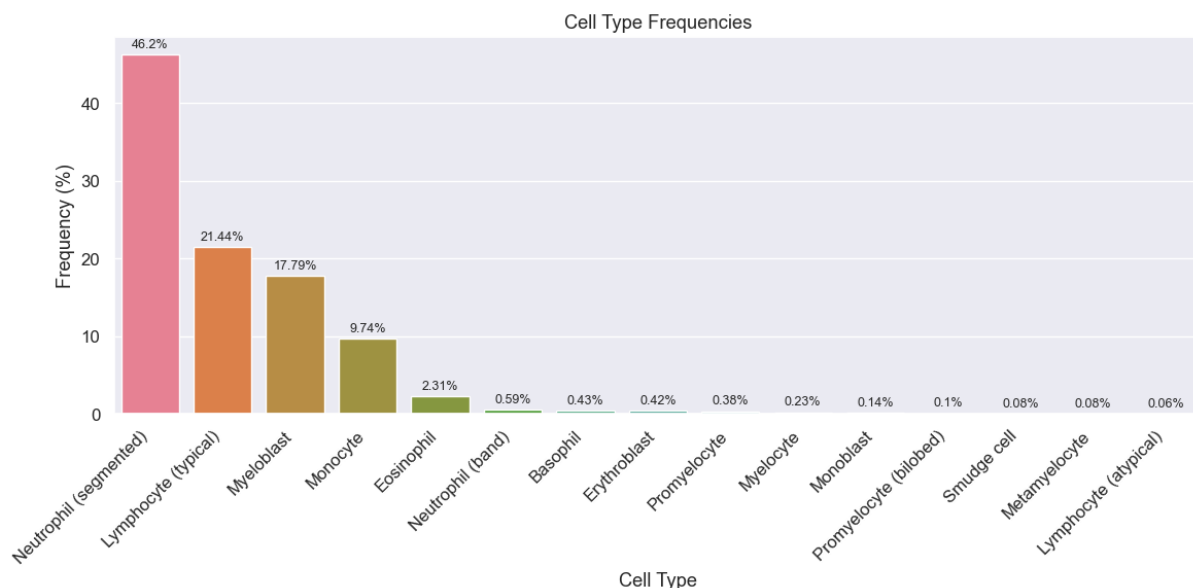
outliers occurring predominantly in the same tail of the distribution. A pragmatic approach would be to proceed with training on the complete dataset and only investigate outlier filtering if specific classification problems emerge during model evaluation.

2.2 The Cancer Imaging Archive (TCIA) AML Dataset

The second dataset contains 18,365 single-cell images extracted from peripheral blood smears of 200 individuals: 100 patients diagnosed with various subtypes of acute myeloid leukemia (AML) and 100 patients exhibiting no morphological features of hematological malignancies [Matek et al., 2019].

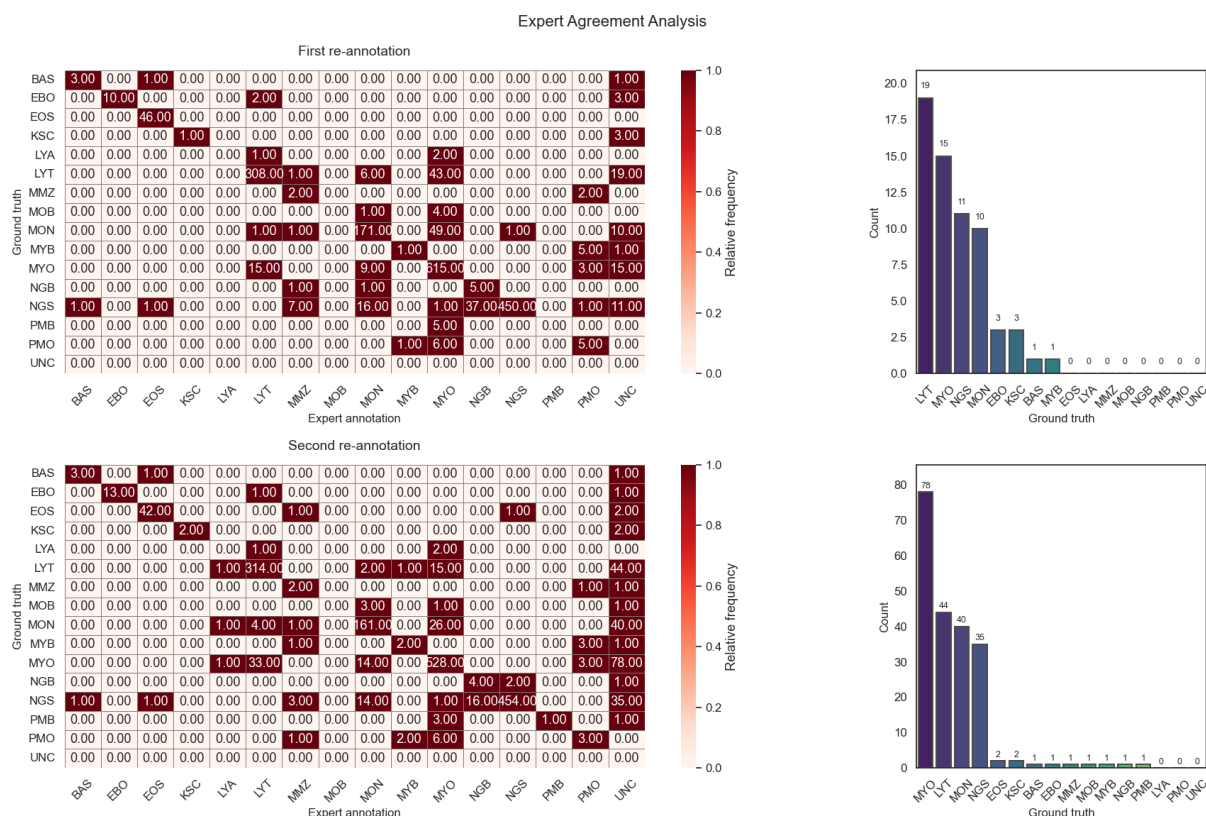
Each cropped image represents a single white blood cell and was assigned a label according to clinically relevant cytomorphological categories used in hematology diagnostics. The dataset includes mature leukocytes, immature granulocyte precursors, and pathological cell types such as myeloblasts, which are crucial for AML diagnosis. The images reflect real clinical variability, including staining differences, background artifacts, and class imbalance, making the dataset particularly suitable for developing robust computer vision models for blood cell classification and leukemia screening tasks.

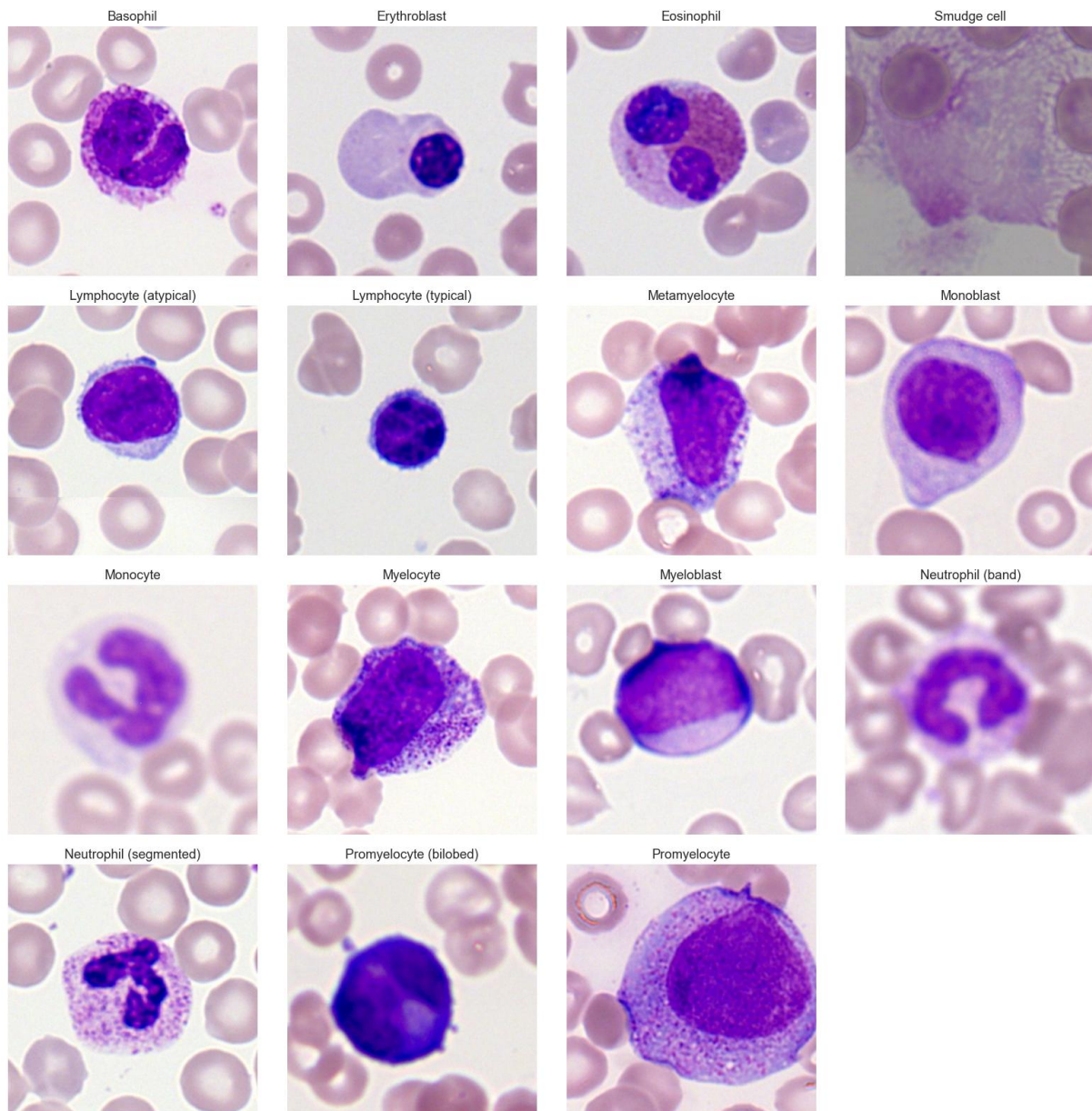
index	cell_type	count	frequency
0	Neutrophil (segmented)	8484	46.20
1	Lymphocyte (typical)	3937	21.44
2	Myeloblast	3268	17.79
3	Monocyte	1789	9.74
4	Eosinophil	424	2.31
5	Neutrophil (band)	109	0.59
6	Basophil	79	0.43
7	Erythroblast	78	0.42
8	Promyelocyte	70	0.38
9	Myelocyte	42	0.23
10	Monoblast	26	0.14
11	Promyelocyte (bilobed)	18	0.10
12	Smudge cell	15	0.08
13	Metamyelocyte	15	0.08
14	Lymphocyte (atypical)	11	0.06



A subset of 1,905 single-cell images was re-annotated by a same examiner (using only cropped cell images, without smear context). The re-annotation was performed twice, with an 11 months gap, allowing assessment of both inter- and intra-examiner agreement.

The examiner could label images as “unclear” (63 images in the first pass, 208 in the second), reflecting uncertainty in some cases. For clearly assigned labels, agreement with the gold standard was excellent (Cohen’s $\kappa = 0.84$ and 0.87). thus to mention that some classification challenges are inherent to human cytomorphology rather than model limitations.

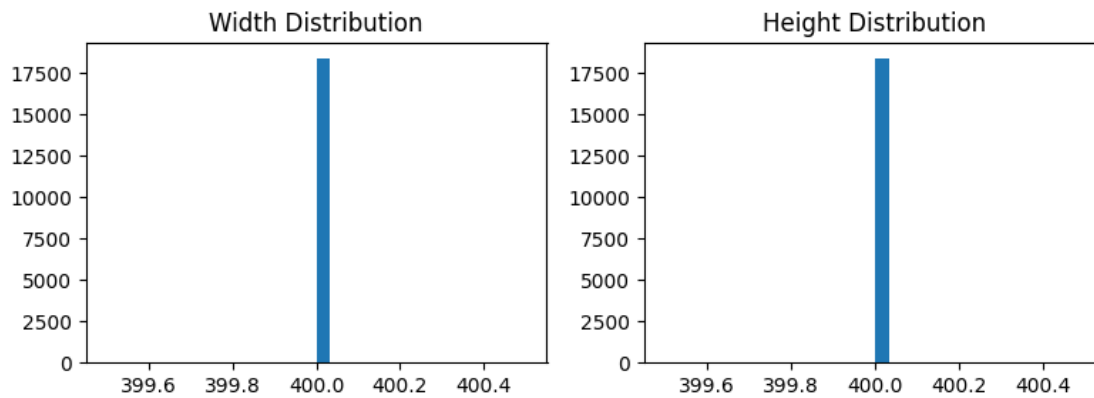




2.2.1 Image Resolution Distribution

The image resolution analysis shows that all leukemia cell images have a uniform resolution of 400×400 pixels, with a total of 18,365 images sharing the same dimensions. This high level of consistency indicates that the dataset was created under standardized imaging conditions.

Because all images have identical resolutions, no additional resizing is required to harmonize image sizes across the dataset. This simplifies the preprocessing pipeline and helps prevent potential information loss or distortions caused by rescaling. Overall, the uniform resolution contributes to a cleaner and more reliable dataset for blood cell classification tasks.



2.2.2 Data Quality Assessment

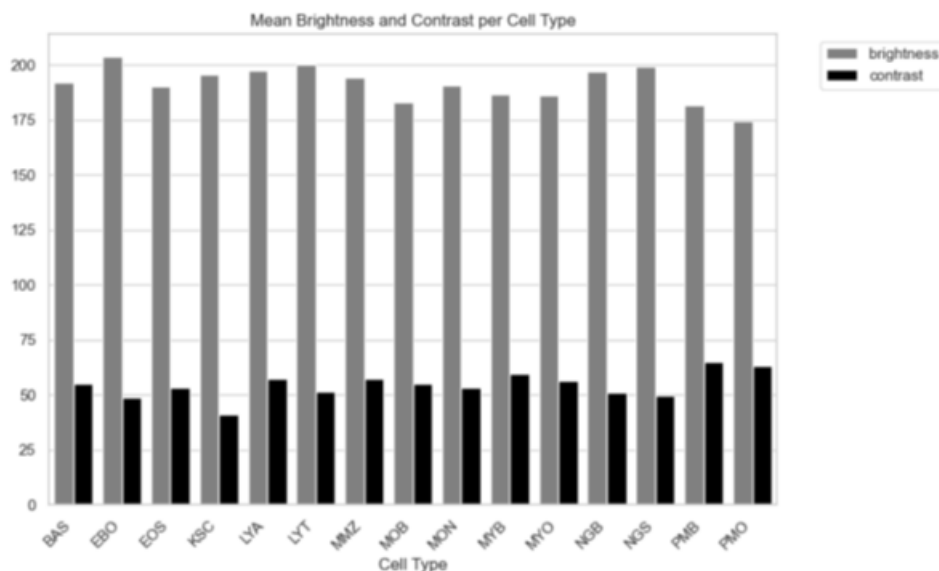
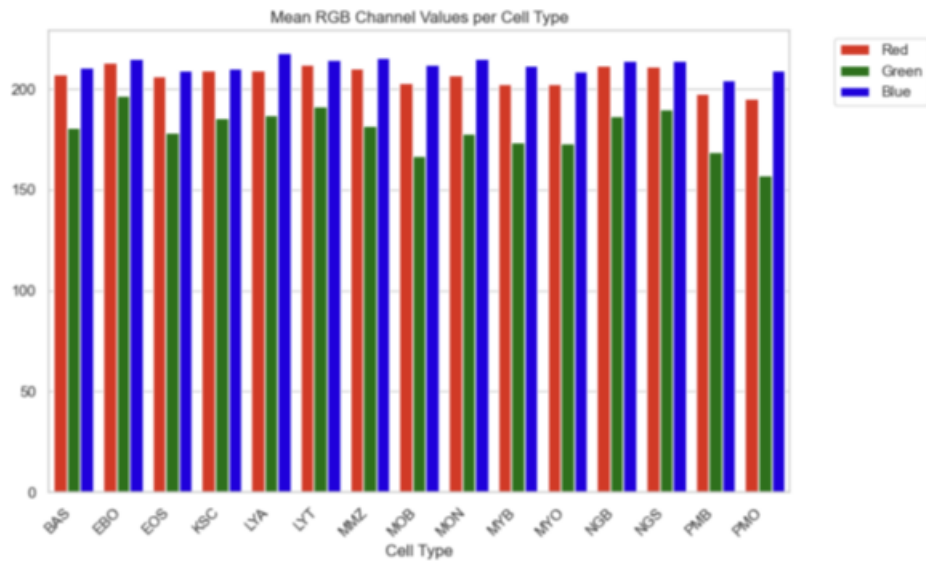
A data quality analysis was conducted on the leukemia dataset to verify image integrity and label consistency. In total, 18,365 images were analyzed, and all images were successfully loaded without errors. No corrupted or unreadable image files were detected during the analysis.

All image file paths were valid and accessible, and no missing labels were found. The dataset contains 15 distinct leukemia-related blood cell types, represented by consistent class labels across the folder structure.

Overall, the leukemia dataset is clean, well-organized, and fully labeled, making it suitable for supervised learning and further blood cell classification tasks.

2.2.3 RGB and Grayscale Intensity Analysis

The mean RGB channel values show consistent patterns, with red values clustering around 200-210, green at 155-195, and blue at 205-220. The red channel dominance observed in the first dataset is less pronounced here, with blue channel intensities elevated. The grayscale brightness analysis reveals moderate variation (175-205) across cell types. PMO shows notably lower brightness values (< 175), while EBO exhibits higher brightness (> 200). Contrast values are relatively uniform (approximately 40-60), with KSC displaying the lowest contrast (~ 40) and PMO and PMB showing elevated levels (~ 65). Overall, the dataset displays greater variability in brightness and color profiles compared to the first dataset, which may provide more discriminative features for CNN classification, though it could also indicate greater imaging heterogeneity.



2.2.4 Distribution Analysis with Boxplots

The boxplot visualizations reveal substantial variability and extensive outlier patterns. MYO exhibits an exceptionally wide brightness range with outliers extending from approximately 115 to 140, indicating severe imaging inconsistencies. KSC, LYT and MON also display considerable spread with low-intensity outliers. The contrast distribution shows that LYT has an extremely broad range spanning from approximately 30 to over 110. For the RGB channels, all three exhibit extensive outlier populations. The prevalence and severity of outliers is substantially greater than in the first dataset and suggests the need for quality control measures before CNN training.

The analysis reveals that this dataset exhibits significantly greater heterogeneity and imaging variability compared to the first dataset, with several classes (particularly KSC, MYO, and PMO) showing extensive outlier populations. The extreme outliers observed - especially in MYO with brightness values ranging from ~115 to over 210 - may indicate potential data

quality issues. From a pragmatic perspective, CNN training can proceed with the complete dataset initially, but if classification problems emerge for these outlier-prone classes, implementing quality control filtering (e.g., removing images beyond $1.5 \times \text{IQR}$) could improve model performance.

3. Merging

3.1 Merging Mendeley and AML datasets

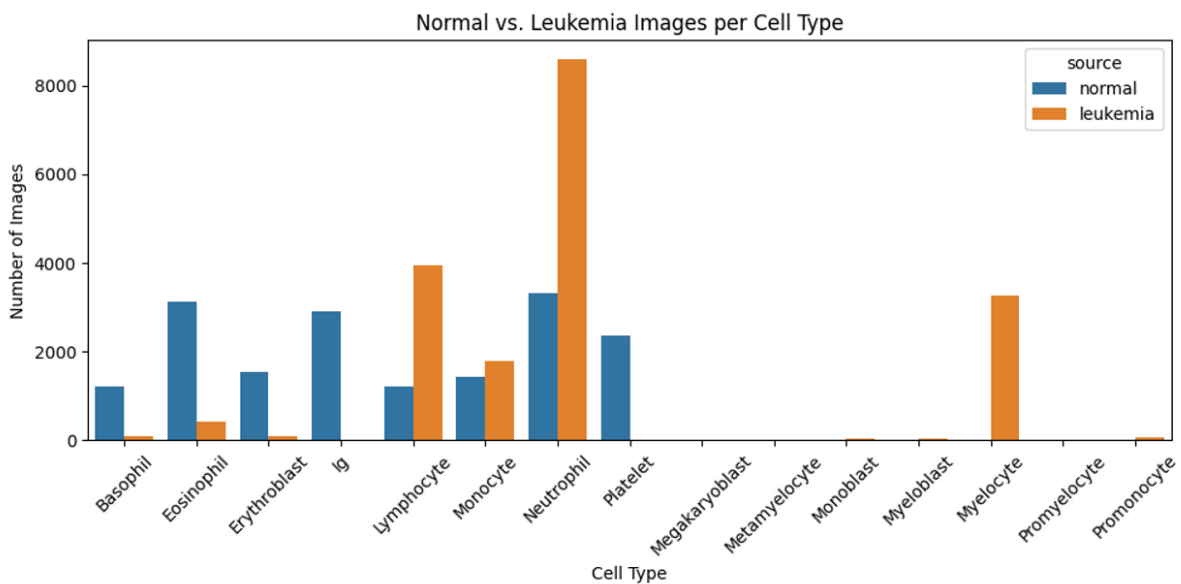
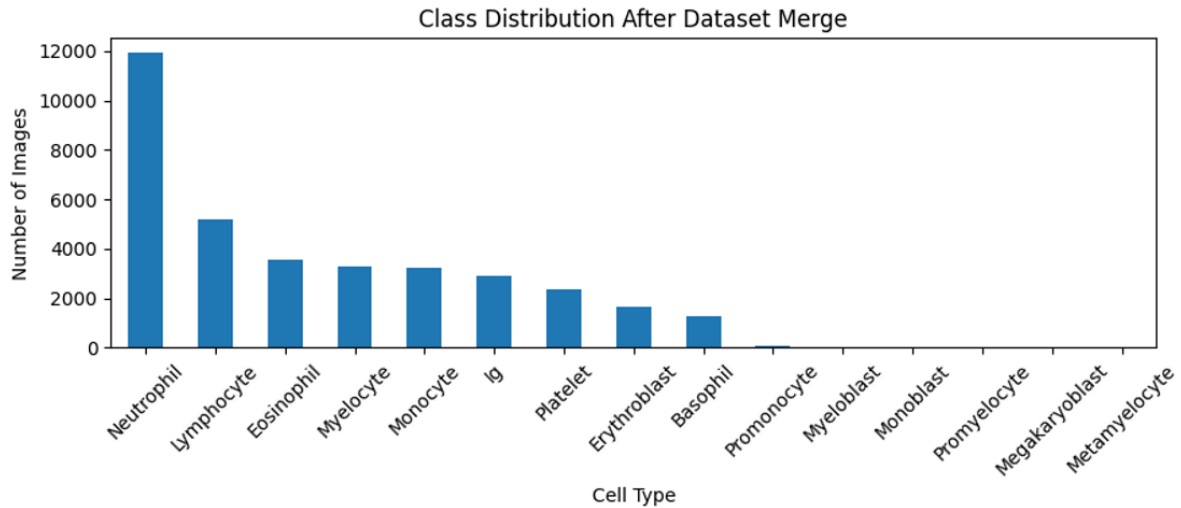
The normal blood cell dataset and the leukemia blood cell dataset were merged to create a more realistic and useful dataset. In real blood samples, normal and abnormal cells appear together, not separately. By merging both datasets, the data better reflects real clinical conditions. The merge also makes it possible to analyze and compare normal cells and leukemia-related cells within a single dataset instead of treating them as two independent problems.

3.2 Advantages of merging

One important advantage is that the merged dataset contains more data and more variation. This helps machine learning models learn more general patterns and reduces the risk of overfitting to only one dataset. Another advantage is practical relevance. Since normal and leukemia cells coexist in real blood samples, a merged dataset is closer to real-world situations and therefore more meaningful for medical applications.

Merging the datasets also allows different types of classification tasks. For example, the data can be used to classify detailed cell types, or to perform simpler tasks such as distinguishing between mature and immature cells.

Finally, the merge makes it easier to compare similar cell types from normal and leukemia data, helping to better understand how disease affects cell appearance.



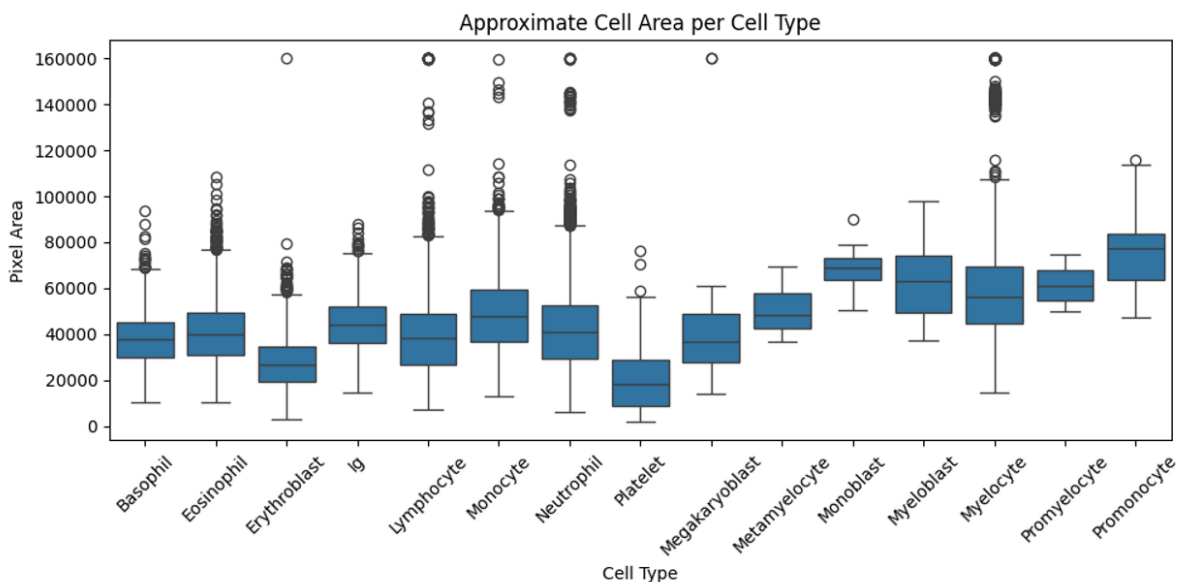
3.3 Disadvantages and challenges

A major challenge of merging the datasets is class imbalance. Some leukemia cell types appear much more frequently than others, which can bias the model if not handled carefully. Another issue is uncertain labels, especially for immature cells. These cells change gradually during maturation, so it is often difficult to clearly distinguish one stage from another. This means that the labels are not always perfectly accurate. In addition, the two datasets may differ in image quality or staining methods, which can introduce unwanted differences that are unrelated to the actual cell types.

3.4 Why merging useful

Even with these challenges, merging the datasets is still useful and commonly done in research. As long as the limitations are clearly explained and the results are interpreted carefully, the merged dataset provides valuable insights.

The goal of merging is not to simplify the data, but to better represent the complexity of real medical images.



3.5 Summary

In summary, merging normal and leukemia blood cell datasets:

- increases data diversity,
- better reflects real blood samples,
- supports different classification tasks,
- but requires careful handling of imbalanced and uncertain labels.

Overall, the merged dataset forms a solid basis for further analysis and model development.